

Solution exercise set 5

Solutions to the problems from the book are found at the end.

Problem 1:

We want to estimate: $\int_1^\infty g(x)dx = \int_1^\infty \frac{e^{-x}}{1+x^2}dx$

Some notes on importance sampling in general:

- We want to estimate the integral $\theta = \int_a^b g(x)dx$; (a, b may be $\pm\infty$).
- Let $f(x)$ be a density function such that $f(x) > 0$ for $a \leq x \leq b$, and let $I_{[a,b]}(x)$ be the indicator function; $I_{[a,b]}(x) = 1$ for $a \leq x \leq b$, zero else.
- We then have for X distributed according to $f(x)$:

$$E\left\{I_{[a,b]}(X) \cdot \frac{g(X)}{f(X)}\right\} = \int I_{[a,b]}(x) \frac{g(x)}{f(x)} f(x) dx = \int_a^b g(x) dx = \theta$$

- $f(x)$ should be close to $g(x)$ so that $\frac{g(x)}{f(x)}$ varies as little as possible for $a \leq x \leq b$.
- $f(x)$ is called the importance function.

Algorithm:

1. Generate $X_1, X_2, \dots, X_n$ from the density $f(x)$.
2. Let $Y_i = I_{[a,b]}(X_i) \cdot \frac{g(X_i)}{f(X_i)}$. We don't know the distribution of Y_i , but since $E(Y_i) = \theta$, the average of $Y_1, Y_2, \dots, Y_n$, $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$, estimates the integral θ .

Since the integral now extends over $[1, \infty)$ and $g(x) > 0$ on $[1, \infty)$, only $f_1(x) = e^{-x}$ and $f_2(x) = \frac{1}{\pi(1+x^2)}$ may be used.

BUT, we may exploit the change of variable $x = 1/y$ to obtain:

$$\int_1^\infty \frac{e^{-x}}{1+x^2} dx = \int_0^1 \frac{e^{-1/y}}{1+y^2} dy$$

Then we may use all the five importance functions from example 6.11.

f_2 : $f_2(x) = \frac{1}{\pi(1+x^2)}$ is the standard Cauchy density. Cdf: $F(x) = \frac{\tan^{-1}(x)}{\pi} + 0.5$.

Inverse, for simulation: $U = \frac{\tan^{-1}(X)}{\pi} + 0.5 \Rightarrow X = \tan\{\pi(U - 0.5)\}$

f_3 : $f_3(x) = \frac{e^{-x}}{1-e^{-1}}$, $0 < x < 1$, truncated exponential density. Cdf: $F(x) = \frac{1 - e^{-x}}{1 - e^{-1}}$.

Inverse, for simulation: $U = \frac{1 - e^{-X}}{1 - e^{-1}} \Rightarrow X = -\ln\{1 - U(1 - e^{-1})\}$

f_4 : $f_4(x) = \frac{4}{\pi(1+x^2)}$, $0 < x < 1$ is a truncated Cauchy density.

Cdf: $F(x) = \frac{4}{\pi} \int_0^x \frac{1}{(1+u^2)} du = \frac{4}{\pi} \tan^{-1}(x)$.

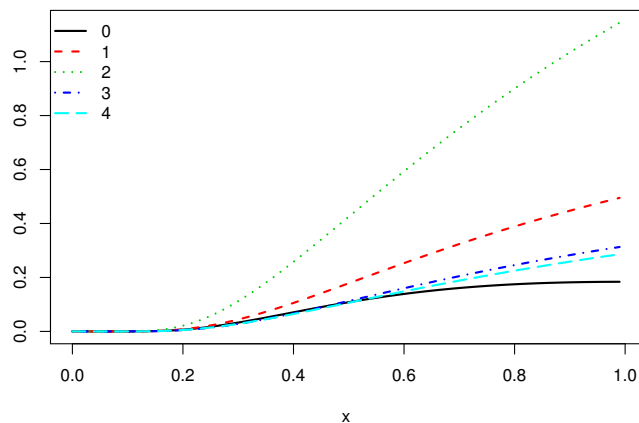
Inverse, for simulation: $U = \frac{4}{\pi} \tan^{-1}(X) \Rightarrow X = \tan(U \frac{\pi}{4})$

Results:

	[,1]	[,2]	[,3]	[,4]	[,5]
theta.hat	0.0973	0.0995	0.0994	0.0953	0.0964
se	0.0702	0.1503	0.2535	0.1011	0.0933

Based on the standard errors (se), f_0 seem to perform best; f_1 and f_2 worst. This also agrees with the results in the figure below, indicating that $g(x)/f(x)$ varies least for $f = f_0, f_4$ and f_3 .

Plot of the functions $\frac{g(x)}{f(x)}$:



Exercises from the book:

Rizzo, exercise 6.6:

$$\theta = \int_0^1 e^x dx = e - 1 = 1.718282$$

Remember that for two random variables X and Y the covariance may be expressed as: $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$. Therefore, if $U \sim U[0, 1]$, then

$$\text{Cov}(e^U, e^{1-U}) = E(\underbrace{e^U e^{1-U}}_e) - \underbrace{E(e^U)}_\theta \underbrace{E(e^{1-U})}_\theta = e - \theta^2 = e - (e - 1)^2,$$

Further since $E(e^{2U}) = \int_0^1 e^{2x} dx = [\frac{1}{2}e^{2x}]_0^1 = \frac{1}{2}(e^2 - 1)$ we get (see also example 5.7):

$$\text{Var}(e^U) = E(e^{2U}) - E(e^U)E(e^U) = \frac{e^2 - 1}{2} - (e - 1)^2$$

$$\begin{aligned} \text{Var}(e^U + e^{1-U}) &= \text{Var}(e^U) + \text{Var}(e^{1-U}) + 2\text{Cov}(e^U, e^{1-U}) = 2\text{Var}(e^U) + 2\text{Cov}(e^U, e^{1-U}) \\ &= e^2 - 1 - 2(e - 1)^2 + 2(e - \theta^2) = e^2 - 1 - 4(e - 1)^2 + 2e \end{aligned}$$

With $\hat{\theta}_1$ the simple MC estimator based on $U_1, \dots, U_m$, i.e. $\hat{\theta}_1 = \frac{1}{m} \sum_{i=1}^m e^{U_i}$, we have

$$\text{Var}(\hat{\theta}_1) = \frac{\text{Var}(e^U)}{m}.$$

With $\hat{\theta}_2 = \frac{1}{m} \sum_{i=1}^{m/2} \{e^{U_i} + e^{1-U_i}\}$, the antithetic MC estimator based on the $m/2$ pairs $U_1, 1 - U_1, \dots, U_{m/2}, 1 - U_{m/2}$, we have

$$\text{Var}(\hat{\theta}_2) = \left(\frac{1}{m}\right)^2 \frac{m}{2} \text{Var}(e^U + e^{1-U}) = \frac{1}{2} \frac{2\text{Var}(e^U) + 2\text{Cov}(e^U, e^{1-U})}{m} = \frac{\text{Var}(e^U) + \text{Cov}(e^U, e^{1-U})}{m}$$

Thus, the variance reduction is:

$$\frac{\text{Var}(\hat{\theta}_1) - \text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)} = \frac{-\text{Cov}(e^U, e^{1-U})}{\text{Var}(e^U)} = \frac{-e + (e - 1)^2}{\frac{e^2 - 1}{2} - (e - 1)^2} = \frac{0.2342}{0.2420} = 0.96767.$$

I.e. the variance reduction is 96.77% .

Rizzo, exercise 6.7:

Some general notes on the MC antithetic variate approach:

- Integral to estimate: $\theta = \int_a^b g(x)dx$.
- If U is uniformly distributed on $[a, b]$ ($U \sim U[a, b]$), then $E\{g(U)\} = \int_a^b g(x) \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \theta$.

And also: $E\{g(1-U)\} = \int_a^b g(x) \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \theta$.

- If $g(x)$ is monotone for $a \leq x \leq b$, we can use the MC antithetic variate approach.
 - Put $m = n/2$; for $U_1, U_2, \dots, U_m$ where $U_i \sim U[a, b]$, the MC antithetic estimate is:

$$\hat{\theta} = \frac{b-a}{2m} \sum_{i=1}^m \{g(U_i) + g(1-U_i)\}$$

- Estimate of the corresponding standard error, SE:

* Let $Y_i = g(U_i)$ and $Y'_i = g(1-U_i)$. Then $\hat{\theta} = \frac{b-a}{2m} \sum_{i=1}^m (Y_i + Y'_i)$, and

* $\text{Var}(\hat{\theta}) = \left(\frac{b-a}{2m}\right)^2 \sum_{i=1}^m \text{Var}(Y_i + Y'_i) = \frac{(b-a)^2 \text{Var}(Y_i + Y'_i)}{4m}$, and the estimated standard error is the square root of this.

* R code for SE: `((b-a)*sd(g.u+g.v)/sqrt(4*m))` when `g.u <- g(u); g.v <- g(1-u)`

Simulation estimate of variances (`n.replic = 10000`):

Simple MC estimator, $\text{Var}(\hat{\theta}_1)$: 2.467e-05

Antithetic variable MC estimator, $\text{Var}(\hat{\theta}_2)$: 7.8e-07

Variance reduction: $100 \frac{2.467 \cdot 10^{-5} - 7.8 \cdot 10^{-7}}{2.467 \cdot 10^{-5}} = 96.8\%$

Rizzo, exercise 6.10:

$$\theta = \int_0^1 \frac{e^{-x}}{1+x^2} dx$$

If $U \sim U[0, 1]$ and $g(U) = \frac{e^{-U}}{1+U^2}$ then $E\{g(U)\} = E\{g(1-U)\} = \int_0^1 \frac{e^{-x}}{1+x^2} dx$.

$$\text{We have } g'(x) = \frac{-e^{-x}(1+x^2) - e^{-x}2x}{(1+x^2)^2} = \frac{-e^{-x}(1+x^2-2x)}{(1+x^2)^2} = \frac{-e^{-x}(1-x)^2}{(1+x^2)^2},$$

i.e. negative for $0 < x < 1$ (all x ;-)) and $g(x)$ is monotone and thus MC with antithetic variables may be used.

(Estimated correlation between the variables $g(U)$ and $g(1-U)$: -0.964.)

Estimates (n.replic = 10⁴):

Estimator	estimate	estimated SE
Simple MC estimator	0.5267685	0.002461
Antithetic variable MC estimator	0.5246464	0.000464

$$\text{Variance reduction: } 100 \frac{0.002461^2 - 0.000464^2}{0.002461^2} = 96.4\%$$

Rizzo, exercise 6.12:

$\theta = \int_A g(x)dx = \int_A \frac{g(x)}{f(x)} f(x)dx$; $X_1, \dots, X_n$: i.i.d. with density $f(x)$. Let $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \frac{g(X_i)}{f(X_i)}$.

We have:

$$\text{Var}(\hat{\theta}) = \text{Var}\left\{\frac{1}{n} \sum_{i=1}^n \frac{g(X_i)}{f(X_i)}\right\} = \frac{\text{Var}\left\{\frac{g(X_i)}{f(X_i)}\right\}}{n} = n^{-1} \left[E\left\{\left(\frac{g(X_i)}{f(X_i)}\right)^2\right\} - \left(E\left\{\frac{g(X_i)}{f(X_i)}\right\}\right)^2 \right]. \text{ We have,}$$

$E\left\{\frac{g(X_i)}{f(X_i)}\right\} = \theta$, and if $|\frac{g(x)}{f(x)}| \leq M$, we have:

$$E\left\{\left(\frac{g(X_i)}{f(X_i)}\right)^2\right\} = \int_A \left(\frac{g(x)}{f(x)}\right)^2 f(x)dx \leq \int_A (M)^2 f(x)dx = M^2 < \infty.$$